SAINT MICHAEL COLLEGE OF CARAGA

THE IMPACT OF ADOPTING A POINT OF SALE SYSTEM ON OPERATIONAL EFFECTIVENESS AND EFFICIENCY.

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The increasing shift toward digital financial technologies has transformed the way businesses manage transactions, with Point-of-Sale (POS) systems and electronic payment platforms becoming essential tools for operational efficiency. As cash usage declines and digital infrastructures expand, organizations worldwide—including food and beverage (F&B) establishments—are relying more heavily on POS technologies to streamline customer service, improve financial transparency, and optimize inventory and data management processes. In the municipalities of Buenavista, Nasipit, and Carmen, the adoption of the Alliance OS POS system reflects this growing transition toward digitalization, offering businesses faster service, real-time business information, and improved accuracy in financial operations.

Globally, recent studies highlight the rapid evolution of e-wallets and digital payments, emphasizing their role in reshaping financial behavior and improving transactional experiences. In Indonesia, internet growth and financial technology have spurred widespread adoption of e-wallets, although research continues to focus mostly on benefits rather than potential risks (Underdown & Tamara, 2025). Internationally, POS systems have also advanced through big data integration, enabling businesses to monitor inventory and restocking more efficiently (Shaba, 2024) . In Nigeria, POS service providers have positively influenced agricultural markets by increasing access to financial services for farmers (Dewi et al., 2021; Kholod et al., 2024; Lesiak et al., 2024; SHARKAWI, 2024). Studies also reveal that paperless billing increases payment speed, security, and efficiency (Faruki et al., 2023; Swathi et al., 2020). Furthermore, electronic financial systems—supported by the internet and mobile technologies—have become crucial for enhancing affordability and accessibility (Mahathir et al., 2025; Twaliwi et al., 2023). These global

insights illustrate how digital payment infrastructures support commercial growth, operational efficiency, and customer satisfaction worldwide.

In the Philippines, digital payment adoption is increasing but remains partial, influenced by policy improvements, digital infrastructure, and user trust (Calderon, 2025; Glindro et al., 2024; Mae Jacob, 2024; Sentral Ng Pilipinas, 2024) . Mobile wallets such as GCash remain the most widely used cashless payment method due to convenience and security, followed by debit/credit cards and online banking, all of which contribute to the growing trust in digital financial systems (Borja et al., 2024; Hernandez et al., 2022; Mathew Jun P. Mariani, 2022; Ramos de Luna et al., 2024) . Institutions such as Mariano Marcos State University have implemented self-service kiosks to reduce queues and enhance customer experience, reflecting local adaptation to digital technologies (Calderon, 2025; Marie Hilay et al., 2021; Timoteo, 2022) . Despite this progress, several studies highlight recurring issues such as inventory mismatches and inaccurate sales reporting, reinforcing the need for stronger internal controls and dependable POS systems in businesses. These insights bridge the global technological context with the Philippine setting, emphasizing the importance of improving POS efficiency and accuracy in the country's food and beverage sector.

Given the increasing adoption of digital payment systems in the Philippines and the challenges many businesses still face in accuracy, inventory control, and service quality, it becomes essential to examine the effectiveness of POS systems in local food and beverage establishments. This study is necessary to address operational issues such as inventory mismatches, service delays, and errors in sales reporting that affect profitability and customer satisfaction. Specifically, the research aims to: (1) identify the demographic profile of respondents based on age, sex, educational attainment, and work experience; (2) assess the

effectiveness of POS systems in improving order accuracy, service speed, and customer satisfaction; and (3) determine how many hospitality businesses in Buenavista, Nasipit, and Carmen use POS systems. By evaluating these factors, the study seeks to determine the extent to which the Alliance OS POS system enhances operational efficiency and decision-making in F&B establishments.

The findings of this study will be valuable to multiple stakeholders within the food and beverage industry. Restaurant owners and managers can use the results to make informed decisions about acquiring or upgrading POS systems to reduce human error, streamline operations, and increase profitability. Employees will benefit from more efficient workflows and reduced workload related to manual tracking and order processing, potentially improving service quality and job satisfaction. Customers will experience faster, more accurate service, contributing to higher satisfaction and loyalty. The study also offers practical implications, such as guiding establishments in improving operational policies, upgrading digital infrastructure, and enhancing internal controls. If the study produces a proposed output—such as recommendations for POS system enhancement or a model for efficient POS usage—it may further support businesses in adopting modernized, data-driven management practices.

1.2 Theoretical Framework

This study is anchored on the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2), which explains how individuals and organizations accept and use new technologies. UTAUT2 proposes that technology adoption is shaped by several key determinants, performance expectancy, effort expectancy, social influence, facilitating conditions, price value and habit. These constructs help clarify why certain establishments adopt technologies like POS systems more readily and benefit from them more effectively. In the context of this research, the

establishment's profile variables (age, sex, educational attainment and employment status), along with system-related conditions (internet availability, location, consistency, accuracy and security), function as the independent variables that influence how UTAUT2 determinants shape the adoption of a POS system. Establishments that perceive the POS system as useful, easy to use, socially supported and adequately facilitated are more likely to adopt and integrate it into daily operations.

Once the POS system is adopted, it generates several operational outcomes that align with the dependent variables of the study, including improved awareness, convenience, security and safety, trust, reduced reliance on cash, social pressure to modernize and increased customer spending. These outcomes directly contribute to enhanced operational effectiveness and efficiency by streamlining transaction processes, reducing errors, improving data accuracy, strengthening financial control and enabling faster, more reliable customer service. Through this theoretical lens, POS adoption serves as the mediating mechanism that connects establishment characteristics and system conditions to measurable operational gains.

This theoretical framework is significant because it provides a structured explanation of how and why POS systems influence organizational performance. It not only identifies the factors that drive adoption but also clarifies the pathways through which the technology generates operational benefits. By grounding the study in UTAUT2, the research ensures that the relationship between the independent variables, POS adoption and operational outcomes is logically connected, theoretically supported and aligned with existing technology adoption literature. This enables a deeper and more systematic examination of the impact of POS systems on operational effectiveness and efficiency.

1.3 Review of Related Literature

The Point-of-Sale (POS) systems play a critical role in improving operational effectiveness and efficiency across retail, hospitality, and service industries. Studies highlight that POS systems significantly speed up transactions, reduce manual workload, and minimize human error, resulting in faster service delivery and higher customer satisfaction. Automated sales recording and real-time inventory tracking enhance internal control, prevent stockouts, and support accurate forecasting and decision-making, thereby increasing productivity and reducing losses (Alt, 2021; Azhari & Sutarman, 2024; Widjaja et al., 2021).

Research further emphasizes that POS analytics strengthen managerial functions such as supplier coordination, demand planning, promotional strategies, and loss prevention, ultimately improving profitability. Modern POS features—such as digital payments, e-receipts, and loyalty programs—also enhance customer experience and retention (Hassan et al., 2021; Jayson Diaz1 et al., 2022; Naik et al., 2024).

However, several studies note challenges in adoption, especially among SMEs, including high upfront costs, maintenance expenses, financial risks, and concerns regarding data security, privacy, and fraud (Akinsuru Adedoyin & Oreoluwa Ojo, 2024; Mamudu Zebedee Udo, 2021; Rubyet Hossain; et al., 2020) . Limited technological infrastructure, lack of digital literacy, and insufficient organizational readiness also hinder effective use of POS data for operational improvement. Despite these barriers, the overall evidence demonstrates that adopting POS systems leads to more streamlined operations, better resource management, and enhanced service efficiency, making it a vital tool for businesses aiming to remain competitive and operationally effective.

Establishments' profile

Demographic factors such as age, gender, employment status, and educational attainment play a significant role in shaping individual behaviors and adoption patterns in various contexts, including technology use and financial practices. Age often influences familiarity and comfort with new technologies, with younger individuals typically demonstrating higher adaptability, while gender can affect preferences and engagement levels in adopting innovations. Employment status determines access to financial resources and the perceived necessity of adopting new systems, whereas educational attainment contributes to awareness, understanding, and confidence in using technological solutions. These variables collectively provide a comprehensive framework for understanding behavioral intentions and adoption patterns, as highlighted in studies examining technology adoption, including e-wallet usage (Rahi et al., 2019).

Identified perceived benefits, effort expectancy, social influence, trust, awareness, and demographic variables—including age, gender, and educational attainment—as significant predictors of technology adoption. These findings highlight the critical role of both psychological and demographic factors in shaping organizational decisions to implement technological innovations. In the context of point of sales (POS) systems, these insights suggest that understanding these determinants is vital for evaluating how the adoption of such systems can enhance operational processes, optimize transaction management, and improve overall organizational efficiency. By considering both individual and contextual factors, organizations can better facilitate the successful integration of POS technologies, thereby maximizing their impact on operational effectiveness Gholami et al. (2010).

The store crew, acting as cashier-level staff, are the primary front-line users of the POS system, operating it during daily transactions and sales activities. In addition, managerial

personnel use the POS system for supervisory functions, such as monitoring operations, reviewing inventory data, and making management decisions. This indicates that the POS system is role-based, with its use concentrated specifically on cashiers and managers within the establishment (Bienvenido, 2023)

Customers' perception

Research on customers' perception of Point-of-Sale (POS) systems in Nigeria highlights the importance of various technological, operational, and service-related variables that shape user experience within cashless and e-payment environments. Key factors frequently examined in the literature include Internet quality, which influences connectivity and transaction speed; Completeness and Consistency, which relate to the reliability and uniformity of POS functions; Observability, referring to the visibility and clarity of transaction processes; and Accuracy, which ensures error-free operations during payments. Additional variables such as Location, Pricing, and Security are also recognized as important contextual and structural elements that affect customer interaction with POS machines. Together, these variables provide a comprehensive framework for understanding how customers form perceptions of POS technology and how these perceptions can guide improvements in Nigeria's cashless and electronic payment systems (Jimoh et al., 2023).

This chapter reviews literature on the adoption and impact of POS systems on operational effectiveness and efficiency, focusing on key factors influencing usage.

Age significantly affects digital payment adoption. Younger users tend to embrace POS systems for their convenience and speed, while older adults may face challenges due to limited technological knowledge and security concerns. Targeted marketing and user-friendly designs

can improve acceptance across age groups (Al-Qudah, Al-Okaily, Shiyyab, et al., 2024; Franco, 2023a; Jiang & Lau, 2023; Martinez & McAndrews, 2022).

Gender influences the perception and use of POS technologies. Men generally prioritize efficiency and functionality, whereas women focus more on trust, safety, and social approval (Gutierrez-Leefmans & Olaleye, 2021; Malarvizhi et al., 2022; Reynolds et al., 2023).

Educational attainment also plays a crucial role, as highly educated users are more adept at understanding and utilizing digital payment systems, facilitating smoother operations and better adoption rates (Al-Qudah, Al-Okaily, Alqudah, et al., 2024; Franco, 2023b; Hsiao & Tang, 2025).

Employment status affects POS efficiency, with employee training and adaptability influencing operational outcomes. Integration of POS with inventory and workflow systems can enhance speed, accuracy, and demand forecasting, though challenges remain with temporary or less tech-savvy staff (Kholod et al., 2024b; Yoon, 2023).

Awareness is critical, especially in underbanked regions; education, demonstrations, and step-by-step onboarding programs increase adoption and reduce errors (Goel et al., 2020; Isharyani et al., 2024; Ly & Ly, 2024).

Convenience including speed, ease of use, and perceived utility—directly affects satisfaction and continued POS usage. Operational efficiency improves when transactions are faster and more reliable (Eren, 2024; Mamudu Zebedee Udo, 2021; Moncada et al., 2025).

Security and safety concerns, such as fraud and data privacy, remain key barriers to adoption. Implementing technical safeguards and educating users enhance trust and reduce risks (Cinar & Kara, 2023; Rawlings Fayeofori Fiberesima, 2025; Santos & Bacalhau, 2023).

Trust in technology and institutions encourages adoption and long-term use. Strong brands, reliable platforms, and trust-building measures contribute to greater frequency of POS usage (Alrawad et al., 2023; Talwar et al., 2020; Zhang & Kim, 2025).

Social Influence also influences adoption, especially in collectivist cultures, where peer recommendations and societal norms impact usage (Asrol, 2024; Cao et al., 2024; Shah & Asghar, 2023).

Limit of cash movements is accelerated by POS adoption, promoting faster, cashless transactions, better banking visibility, and operational efficiency (Glowka et al., 2023; Islam et al., 2024).

Consumer spending behaviors, leading to higher transaction values and improved business revenue, though inventory and operational management must be carefully addressed (Broekhoff & van der Cruijsen, 2024; Hidayati et al., 2023; Ma et al., 2024).

Location

Location significantly influences the operational effectiveness and efficiency of businesses adopting a Point-of-Sale (POS) system. Kholod et al. (2024) emphasized that integrating POS data with location-tracking technologies, such as RFID, enables businesses to monitor customer movement patterns and transaction points within specific areas, supporting optimized store layouts, strategic product placement, and efficient staff allocation.

Similarly, Ighomereho & Omoyele (2022) highlighted that the reliability, ease of use, and fulfillment of POS systems across different service points are critical for ensuring smooth, secure, and efficient transactions, directly affecting operational performance. These findings indicate that strategic placement of POS terminals within specific locations enhances transaction

efficiency, optimizes resource allocation, and improves overall customer satisfaction, demonstrating the importance of location in operational outcomes.

Internet

Internet connectivity is a vital factor influencing how effectively and efficiently a Point-of-Sale (POS) system operates. Walter (2024) argues that reliable multi-network cellular connectivity (e.g., switching between 4G/5G and Wi-Fi) ensures POS devices remain online even when one network fails, reducing transaction downtime and improving resiliency. Meanwhile, Kayla (2025) explains that slow or unstable broadband can severely delay payment processing, disrupt real-time inventory updates, and cause synchronization errors in cloud-based POS platforms, ultimately degrading both customer experience and operational throughput.

Consistency

Consistency in service delivery is a key factor in determining how well a POS system enhances operational effectiveness and efficiency. Gao & Fan (2021) found that when customers perceive their online and offline experiences to be congruent (i.e., consistent), satisfaction, repurchase intention, and word-of-mouth improve, whereas any discrepancy between channels undermines these outcomes.

In banking contexts, Olukemi Oluyemisi & Abba (2024) noted that the reliability dimension of POS terminal service, closely linked to consistent availability across POS channels, strongly influences customer satisfaction. Thus, ensuring consistent POS service quality across all touchpoints supports smoother operations, reduces customer frustration, and promotes greater operational efficiency.

Accuracy

Accuracy is a critical dimension in how adopting a POS system affects operational effectiveness and efficiency Okofu (2025) developed an enhanced POS system for retail businesses in developing countries that ensures transactions are “reliable, accurate, faster, and secure” by using barcode scanning and a recommender- system architecture, minimizing human error and improving the precision of sales records.

Moreover, Zídková et al. (2025) discusses the digital registration of cash sales via POS systems in public finance, showing that accurate real- time reporting of cash transactions can strengthen fiscal resilience and operational transparency. Together, these studies highlight that accuracy in POS transactions not only reduces errors and discrepancies but also enhances data reliability for decision-making and operational control.

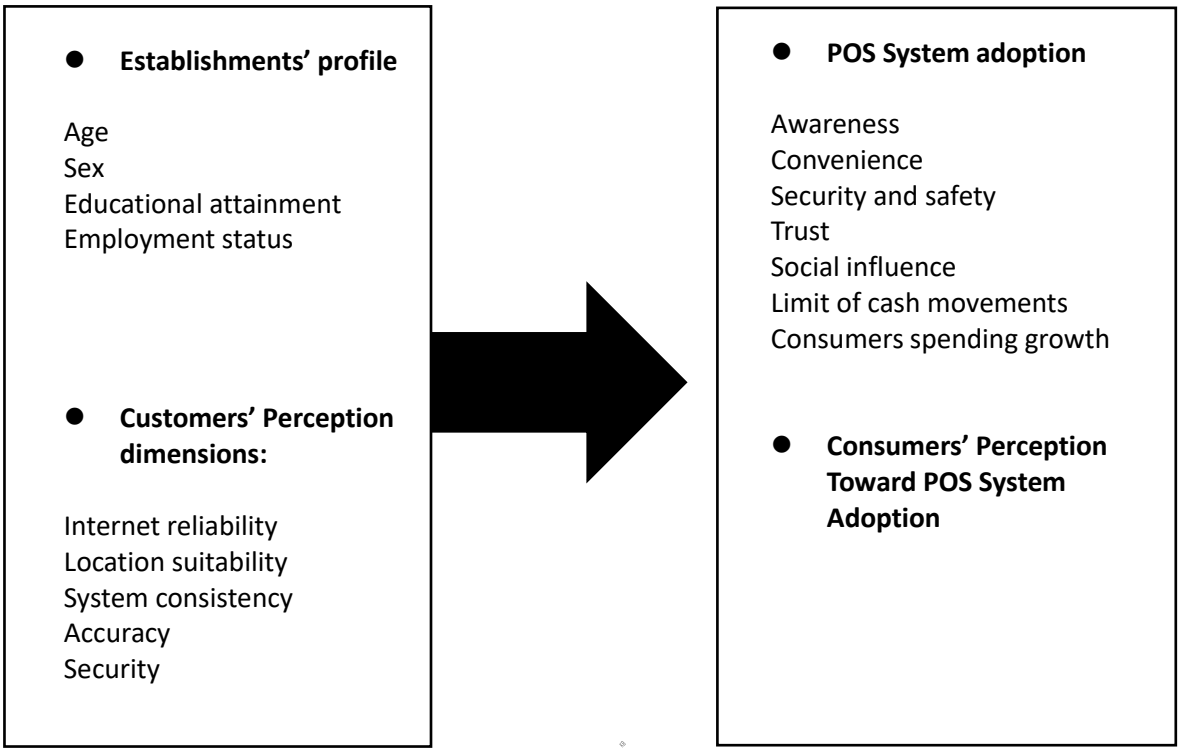
Customers’ perception

Customer perception strongly influences how the adoption of a POS system affects operational effectiveness and efficiency. Röding et al. (2023) classified different types of POS technologies from the customer’s viewpoint, showing that customers’ perceptions of information- oriented POS systems (e.g., how they present product info or transaction details) vary and directly affect their acceptance.

Meanwhile, González-Morera et al. (2023) found that customers visiting Caracol points-of-sale perceive visual merchandising elements positively overall, though aspects such as product distribution, promotions, and ambiance (like music) are seen less favorably, suggesting that POS implementations reinforcing attractive, well-organized store layouts can boost customer satisfaction.

1.4 Schematic Diagram

Diagram 1. Establishments’ profile



The diagram shows how the independent variables relate to the dependent variable. Age, gender, education, and occupation of the respondents constitute the independent variable (IV). It is expected that these factors will influence the respondents' attitudes, intentions, and perceptions regarding the use of POS systems in their businesses or workplaces. And the independent variables represent key factors associated with the adoption of a Point of Sale (POS) system, including Internet connectivity, Location, Consistency, Accuracy, and Security. These factors are expected to influence the overall performance and perception of the system.

The dependent variables (DVs) reveal the significant areas of effectiveness and productivity. They are measured by consumer spending growth, cash use constraints, social impact, convenience, security, trust, and awareness, among others. These factors capture the results and behavior changes from using POS systems, including how users see and trust the

technology, as well as real changes in efficiency, customer satisfaction, and spending habits. And the customers' perception towards POS machines, which reflects the degree to which customers notice and respond to the presence and use of POS systems in business locations. This variable captures the impact of POS adoption on customer experiences and, indirectly, on operational effectiveness and efficiency.

1.5 Statement of the Problem/ Objectives of the Study

Sop 1. What is the profile of the respondents in terms of:

1.1 Age

1.2 Sex

1.3 Educational Attainment

1.4 Employment status

Sop 2. What is the level of impact on POS adoption among establishments in terms of:

2.1 Awareness

2.2 Convenience

2.3 Security and Safety

2.4 Trust

2.5 Social Influence

2.6 Limit of cash movement

2.7 Consumers' spending growth

Sop 3. Is there a significant relationship between the profile of the respondents and the level of impact of adopting a POS?

Sop 4. How do system-related factors such as:

3.1 Internet connectivity

3.2 location, consistency

3.3 accuracy

3.4 security

Affect customers' perception of POS machine usage within establishments?

1.6 Hypothesis

Null Hypothesis 1:

There is no significant relationship between the respondents' profile and system-related factors (internet connectivity, location, consistency, accuracy, and security) and the overall impact of adopting a POS system in terms of awareness, convenience, security and safety, trust, social influence, limit of cash movement, and consumer spending growth.

Hypothesis 1:

There is a significant relationship between the respondents' profile and system-related factors (internet connectivity, location, consistency, accuracy, and security) and the overall impact of adopting a POS system in terms of awareness, convenience, security and safety, trust, social influence, limit of cash movement, and consumer spending growth.

1.7 Scope and Limitations

This study aims to examine the effects of adopting a Point of Sale (POS) system on operational effectiveness and efficiency in the food and beverage (F&B) industry. The research focuses specifically on F&B establishments operating in the municipalities of Buenavista, Nasipit, and Carmen, where the industry has experienced notable growth due to rising local economic activity and increasing consumer demand. These areas were selected because they host a growing number of small to medium-sized restaurants, cafés, and other F&B businesses, making

them suitable sites for studying technology adoption in business operations. Their accessibility and relevance also provide feasibility for accurate data collection.

The study will concentrate on how POS adoption influences several operational dimensions, including transaction accuracy, service speed, inventory control, cost reduction, and customer satisfaction. Respondents will include employees—cashiers, servers, and managers—who interact directly with the POS system in their daily tasks, as well as customers, who do not engage with the system directly but experience its effects through service speed, billing accuracy, and overall service quality. The research will be conducted and analyzed within School Year (SY) 2025–2026.

Despite its defined scope, the study has several limitations. First, it will only cover a selected sample of F&B outlets within the three municipalities, so findings may not be generalizable to other towns, cities, or larger-scale restaurant operations. Second, the respondent groups are limited to employees and customers, which may result in limited representation from business owners or top management. Third, the study will focus solely on operational outcomes of POS adoption and will not explore broader financial or marketing impacts such as revenue growth, brand positioning, or long-term strategic benefits. Additionally, customer responses may be constrained by their limited awareness of the POS system's technical functions, as their feedback will primarily reflect service-related experiences. Finally, external factors such as internet connectivity, power supply stability, vendor-related software updates, and regulatory requirements may influence POS performance, but these factors fall outside the scope of the study.

1.8 Definition of Terms

Accuracy. Refers to the correctness of the POS system in recording, calculating, and processing transactions. It is measured by comparing system records with actual transactions to identify discrepancies in pricing, billing, or inventory.

Age. Refers to the respondent's chronological age (years) as reported in the survey. Age has been considered a demographic variable in this research, affecting merchants' awareness, trust, perceptions of convenience, and adoption of POS systems.

Awareness. Refers to how familiar respondents are with POS systems. It is measured by their self-reported knowledge, recognition of features, and experience using POS transactions.

Consistency. Refers to the ability of the POS system to perform transactions uniformly and predictably without errors. It is measured by tracking transaction success rates, frequency of system errors, or repeated functionality tests over time.

Convenience. Refers to respondents' perceived ease, speed, and convenience of POS transactions in their work or customer experience. Convenience involves quicker checkout, various payment options, and ease of handling transactions.

Customers' Perception Towards POS Machines. Refers to the opinion or attitude of customers regarding the use of POS machines in a business. It is measured through surveys using a Likert scale on statements about convenience, speed, and satisfaction with POS transactions.

Educational Attainment. Refers to the highest level of education a respondent has completed (e.g., high school, college, or postgraduate). It is used in this study as a measure of digital literacy, which can affect how well they adopt and use POS systems.

Employment Status. Refers to the type of work restaurant employees do, such as full-time, part-time, or temporary/contract. It helps assess how their work status affects the POS system's impact on operational efficiency and effectiveness.

Growth in consumer spending. Refers to the perceived or reported growth in customer spending (basket size, purchase frequency, or sales volume) driven by convenience enabled by the POS and lower friction in purchase transactions.

Internet. Refers to the reliability and speed of the internet connection used by the POS system to process transactions. It is measured by the frequency of transaction delays or failures due to internet issues or through ratings from customers and employees on system connectivity.

Limit of Cash Movements. Refers to the level at which the business manages and reduces the number of times it physically handles, stores, or transfers cash in the daily business.

Location. Refers to the physical or virtual accessibility of the POS system within the business environment. It is measured by evaluating the ease of access for customers and employees to POS terminals or by tracking the time taken to complete transactions at different locations.

Operational Effectiveness. Refers to how POS systems help businesses and employees work more accurately and reliably. Examples include improved inventory control, fewer transaction errors, and smarter decision-making.

Security. Refers to the measures implemented to protect the POS system and customer data from unauthorized access, fraud, or breaches. It is measured by the presence of security protocols (e.g., encryption, user authentication), incidence of security breaches, or the perceived safety by customers and staff.

Security and Safety. Refers to how safe respondents feel when using POS systems, especially regarding fraud, mistakes, and data breaches. It is measured by their trust in the system's technical security and their sense of safety while making transactions.

Sex. Refers to the restaurant customer's sex, which is determined by whether the respondent is male or female, as transpired in the demographic profile part of the questionnaire. This variable tests the potential variation in the perceptions of POS awareness, trust, and security.

Social Influence. Refers to the impact of peers, coworkers, managers, and customers on the respondents' use of or dependence on POS systems. It is calculated by the perceived support or normative influence to use and embrace POS technology.

Trust. Refers to how confident restaurant customers feel about using POS systems. It includes trust in the technology, the service providers, and the merchants. It is measured by survey questions about reliability, transparency, and confidence in using POS information.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter will discuss the research design, locale, respondents, and instruments used in the study. It will also explain the ethical standards, data-gathering procedure, and statistical treatments.

2.1 Research Design

The researchers will utilize a quantitative-descriptive correlational research design to find the relationship between Point-of-sale system and Operational Effectiveness and Efficiency. In this design, the study will involve gathering numerical data through survey questionnaires that describe and analyze the perceptions and behaviors of the respondents. The descriptive part will help the researchers in presenting the characteristics of the respondents, while the correlational part will determine if there is a significant relationship between Point-of-sale system and Operational Effectiveness and Efficiency. This design will be appropriate because it will enable the researcher to collect measurable data and find out how Point-of-sale system may Impact customers' satisfaction in Buenavista, Nasipit, and Carmen.

2.2 Research Materials and Methods

Research Instrument

Two research instruments will be used in this study. The first instrument will be a modified survey questionnaire from the study of Oyelami et al. (2020), for the establishments. The factor-related variables will be studied in line with Oyelami et al. (2020), and the research instrument will be divided into two parts. Part I will focus on establishment employees and their demographic profile, collecting information on age, sex, educational attainment, and employment status. Part II will consist of 27 items, with responses on a Likert scale: 4 for Strongly

Agree, 3 for Agree, 2 for Disagree, and 1 for Strongly Disagree. This survey questionnaire will assess how the point-of-sale system will impact restaurant customers. And the second research instrument will be a modified survey questionnaire from the study by Huma et al. (2025) on customers' perception towards POS adoption. The factor-related variables will be studied in line with Jimoh et al. (2023) , and the research instrument will be divided into two parts. Part I will focus on the demographic profile of the respondents by collecting information on age, sex, educational attainment, and employment status. Part II will consist of 27 items, with responses on a Likert scale: 4 for Strongly Agree, 3 for Agree, 2 for Disagree, and 1 for Strongly Disagree. This survey questionnaire will assess how the point-of-sale system will impact restaurant customers.

2.3 Research Environment, Population, and Sample

This study will explore how using a Point of Sale (POS) system affects the efficiency and performance of food and beverage (F&B) businesses. It will focus on companies in Buenavista, Nasipit, and Carmen, looking at how POS systems impact transaction accuracy, service speed, inventory management, cost savings, and customer satisfaction.

These municipalities were chosen because their F&B industries are growing as the local economy strengthens and demand rises. More restaurants, cafes, and other F&B businesses are opening in these areas.

Table 1. Establishments' employee

	POPULATION			SAMPLE					
RESPONDENTS (Employees)	**Male**	**Female**	**Total**	**Male**		**Female**		**Total**	
	F	**F**	**F**	**f**	%	**f**	%	**f**	%
Buenavista									
Roadside		2	2		0.0	2	10.5	2	10.5
Backyard	1		1	1	5.3	0	0.0	1	5.3
Snack break		2	2		0.0	2	10.5	2	10.5
Cafe Aria		1	1		0.0	1	5.3	1	5.3
Aiedee's		1	1		0.0	1	5.3	1	5.3
Nasipit									
Buprs Koi		1	1		0.0	1	5.3	1	5.3
3Js		1	1		0.0	1	5.3	1	5.3
Cafearia		1	1		0.0	1	5.3	1	5.3
Five star	1	1	2	1	5.3	1	5.3	2	10.5
Project Moi		1	1		0.0	1	5.3	1	5.3
Carmen									
Mahigalaon	1	1	2	1	5.3	1	5.3	2	10.5
Painawa		1	1		0.0	1	5.3	1	5.3
Mt. Carmel		1	1		0.0	1	5.3	1	5.3
Zander's food hub		1	1		0.0	1	5.3	1	5.3
Mangroove		1	1		0.0	1	5.3	1	5.3
Total	**3**	**16**	**19**	**3**	**15.9**	**16**	**84.6**	**19**	**100.0**

Table 1 shows the distribution of establishment employees involved in the study, thereby further expanding the number of employees from Buenavista, Nasipit, and Carmen establishments.

Table 2. Customers

Restaurant	POPULATION			SAMPLE					
	Male	Female	Total	Male		Female		Total	
	F	F	F	f	%	f	%	f	%
Buenavista									
Roadside	120	80	200	18	5.5	12	3.7	30	9.2
Backyard	20	30	50	3	0.9	5	1.5	8	2.4
Snack break	250	250	500	38	11.6	38	11.6	76	23.2
Cafe Aria	60	90	150	9	2.8	14	4.3	23	7.0
Aiedee's	50	70	120	8	2.4	11	3.4	19	5.8
Nasipit									
Buprs Koi	40	60	100	6	1.8	9	2.8	15	4.6
3Js	30	40	70	5	1.5	6	1.8	11	3.4
Cafearia	60	90	150	9	2.8	14	4.3	23	7.0
Five star	100	100	200	15	4.6	15	4.6	30	9.2
Project Moi	91	39	130	14	4.3	6	1.8	20	6.1
Carmen									
Mahigalaon	35	35	70	5	1.5	5	1.5	10	3.1
Painawa	40	60	100	6	1.8	9	2.8	15	4.6
Mt. Carmel	60	40	100	9	2.8	6	1.8	15	4.6
Zander's food hub	50	50	100	8	2.4	8	2.4	16	4.9
Mangroove	50	50	100	8	2.4	8	2.4	16	4.9
Total	**1056**	**1084**	**2140**	**161**	**49.2**	**166**	**50.8**	**327**	**100.0**

Table 2 shows the distribution of restaurant customers involved in the study, thereby expanding the number of employees from the Buenavista, Nasipit, and Carmen restaurants.

2.4 Research Data and Analysis Plan

The researchers will use stratified cluster sampling. The groups (strata) will be based on municipality (Buenavista, Nasipit, Carmen) and type of establishment (fast food, full service, kiosk). Customers and establishments' employee will be randomly chosen from each group, with the number of customers and employee reflecting the size of each type of business.

Statistical Treatment

In presenting, analyzing, and interpreting data, the researcher used the following statistical test, namely.

Frequency count. This tool will be used to answer SOP 1 in determining the profile of the respondents.

Weighted Mean. Will use this tool to answer SOP 2 in determining the level of impact among restaurant customers.

Standard deviation. Will be used to determine the level of agreement or the differences among the restaurants' customer opinions. A low standard deviation means that the responses are nearly centered on the average, thus revealing consensus among the participants.

Correlation analysis. We will apply this to find out the connection between the two variables, mainly the respondents' profiles (age, gender, educational level, and job status) and their opinion on the impact of the POS system on different aspects.

2.5 Research Ethical Standards

A. Intellectual Property Rights

This study will uphold intellectual property rights by ensuring proper acknowledgment of all sources and contributions. All data, literature, and research materials utilized in this study will be accurately cited in accordance with academic standards. Respondents' inputs will remain anonymous, and the researchers' own contributions will be clearly distinguished from those of others, ensuring that the intellectual work of all parties is respected and recognized.

B. Informed Consent

Participants, specifically restaurant customers from Buenavista, Nasipit, and Carmen, will be fully informed about the study before participation. Researchers will conduct a brief orientation session explaining the study's objectives, methodology, and potential benefits and risks.

Participants will receive clear instructions on their rights, including the right to decline participation or withdraw at any time without consequence. A written consent form will be provided to ensure participants fully understand their rights and voluntarily agree to take part in the research.

C. Data Privacy and Confidentiality

All personal information, including demographic and identifying data, will be collected strictly with participants' consent and maintained in strict confidentiality. Data will be anonymized to protect participant identities. Only the research team will have access to identification codes, and all collected information will be securely stored and handled to prevent unauthorized access, disclosure, or misuse.

D. Voluntary Participation

Participation in this study is entirely voluntary. Participants will be informed that they may withdraw at any point without facing any negative consequences or penalties. This study involves no animal subjects, ensuring full ethical compliance regarding human-only participation.

E. Minimization of Harm

Researchers will engage with participants respectfully, professionally, and with sensitivity to their well-being. Any potential emotional, psychological, or physical risks will be minimized, and participants will be referred to appropriate support services, such as clinics or the Guidance Office, if necessary. The study will include follow-up measures to confirm that participation did not cause harm or distress.

F. Beneficence

This study aims to contribute valuable insights into the impact of the POS system on the restaurant industry. The knowledge gained will benefit both participants and the wider

community. Upon request, participants will be provided with a summary of the findings to ensure that the research contributes positively to all stakeholders.

G. Justice and Fair Participant Selection

Participant selection will follow fair, unbiased criteria directly aligned with the study's objectives. Individuals from diverse backgrounds will be included to ensure equitable representation. No discrimination will be tolerated based on gender, age, ethnicity, socio-economic status, or other characteristics.

H. Data Integrity and Accuracy

Data will be collected and analyzed using standardized and scientifically validated methods to ensure reliability and accuracy. The research team will formally request access to respondent lists from the relevant municipalities and will critically review all findings to eliminate bias. Every effort will be made to maintain data integrity throughout the research process.

I. Transparency in Reporting

All findings will be reported honestly, accurately, and without any manipulation of data. Academic sources will be properly cited, and AI tools or other technologies will be used responsibly to assist with data analysis and reporting. Any use of such tools will be clearly disclosed to maintain transparency and uphold ethical standards.

J. Use of Patented or Copyrighted Materials

No patented materials will be used in this study. All copyrighted materials will be appropriately cited and utilized in compliance with academic integrity and copyright law. The research will respect the legal and ethical guidelines surrounding intellectual property.

K. Considerations for Human Subjects

Participants will be informed of all potential risks and benefits associated with the study, and their consent will be obtained before any data collection. If participants experience emotional or psychological discomfort during the study, they will be provided with access to appropriate support services to ensure their well-being.

L. Responsible Use of AI and Related Technology

AI and related technologies will be employed responsibly to enhance data analysis, improve accuracy, and support the research process. Use of such tools will be transparent, ethical, and compliant with academic and legal standards, ensuring that technological assistance does not compromise research integrity.

M. Ethical Approval

This study has received ethical clearance from the Institutional Review Board (IRB) of Saint Michael College of Caraga. This approval confirms that the research adheres to all ethical guidelines, including the protection of respondents, maintenance of data privacy, and respect for intellectual property rights.

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